

	Q1–4	Q5	Q6 abc	Q6 de	Q7 a	Q7 b	Q8 a	Q8 b	Total
HE4	/4		/6			/10			/20
HE6				/5			/6		/11
HE7		/1			/2			/6	/9
									/40

Section I

5 marks

Attempt Questions 1 – 5

Allow about 7 minutes.

Use multiple choice answer sheet for questions 1 – 5

1. What is the derivative of $\log_2 x$?

(A) $\frac{1}{x}$

(B) $\frac{1}{2x}$

(C) $\frac{\ln 2}{x}$

(D) $\frac{1}{x \ln 2}$

2. What is the range of the function $f(x) = \log_e(x^2 + e)$?

(A) $y \geq 0$

(B) $y > 0$

(C) $y \geq 1$

(D) $y > 1$

3. Which of the following is the derivative of $\cos^{-1}(e^{-x})$?

(A) $\frac{e^x}{\sqrt{1-e^{2x}}}$

(B) $\frac{-e^x}{\sqrt{1-e^{2x}}}$

(C) $\frac{e^{-x}}{\sqrt{1-e^{-2x}}}$

(D) $\frac{-e^{-x}}{\sqrt{1-e^{-2x}}}$

4. Which expression is equal to $\int \frac{1}{3+x^2} dx$?

(A) $\tan^{-1} \frac{x}{3} + C$

(B) $\tan^{-1} \frac{x}{\sqrt{3}} + C$

(C) $\frac{1}{3} \tan^{-1} \frac{x}{3} + C$

(D) $\frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} + C$

5. What is the value of $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin 4x}$?

(A) 0

(B) $\frac{5}{4}$

(C) $\frac{4}{5}$

(D) Undefined

Section II

35 marks

Attempt Q6 – 8

Allow about 48 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing paper is available.

In Question 6 – 8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (11 Marks)

Use a Separate writing booklet

a) Evaluate $\int_0^1 2e^{1-x} dx$, in exact form. 2

b) Find the gradient of the tangent to the graph $y = \sin^{-1}(3x - 2)$ at $x = \frac{1}{2}$. 2

c) Using the standard integral sheet, or otherwise, differentiate $y = \log_e(\sec x)$. 2

d) Calculate the exact value of $\cos^{-1}\left(\sin\left(-\frac{\pi}{6}\right)\right)$. 2

Show all necessary working.

e) Evaluate $\int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx$, using the substitution $x = \cos \theta$, where $0 < \theta < \pi$. 3

Leave your answer in exact form.

Question 7 (12 Marks)**Use a Separate writing booklet**

a) Prove that $2 \tan^{-1} \frac{1}{5} = \tan^{-1} \frac{5}{12}$. 2

(Note. Using a calculator does not qualify as a proof.)

b) Let $f(x) = 1 - e^{-x^2}$ for $x \leq 0$

(i) Show that the coordinates of the point of inflexion is $\left(-\frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{e}}\right)$. 3

(ii) Explain why $0 \leq f(x) < 1$. 1

(iii) Evaluate $\lim_{x \rightarrow -\infty} f'(x)$. 1

(iv) On a one-third page diagram, sketch the graph of $y = f(x)$. 1

(v) On the same diagram, sketch the inverse function $y = f^{-1}(x)$. 2

(vi) Find the equation of $y = f^{-1}(x)$ noting any restrictions on the domain. 2

Question 8 (12 Marks)**Use a Separate writing booklet**

a) Consider the function $f(x) = 3 \cos^{-1} \frac{x}{2}$

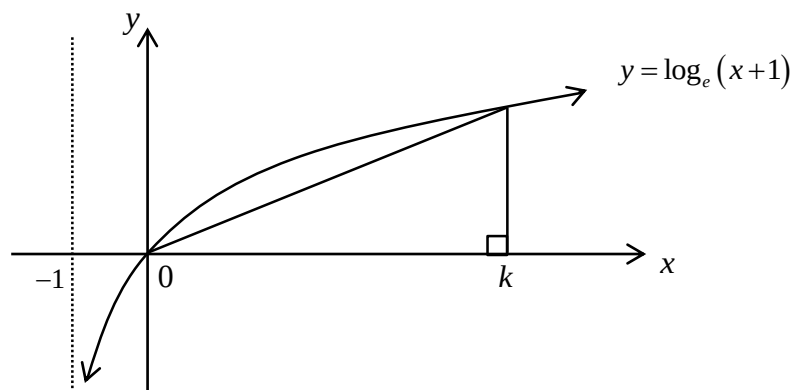
(i) Find the y intercept of the graph of the function $y = f(x)$. 1

(ii) Sketch a graph of $y = f(x)$, showing all important features,
including the intercepts and end points. 2

(iii) Find the exact volume of the solid generated when the region bounded by
the curve, the line $x = 1$ and the x-axis is rotated about the y-axis. 3

(Question 8 continues)

b) The diagram shows the graph of $y = \log_e(x+1)$



- (i) Show that the area of the region bounded by the curve $y = \log_e(x+1)$ and the x -axis for $0 \leq x \leq k$ is given by 3

$$A = (k+1)\log_e(k+1) - k$$

- (ii) By comparing relevant areas in the diagram show that 2

$$\log_e(k+1) > \frac{2k}{k+2} \quad \text{for } k > 0$$

- (iii) Hence, explain why $\frac{\log_e x}{2} > \frac{x-1}{x+1}$ for $x > 1$. 1

End of test

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Answers

Section I

1 (D) 2 (C) 3 (C) 4 (D) 5 (B)

Section II

Question 6

a)

$$\begin{aligned}\int_0^1 2e^{1-x} dx &= -2[e^{1-x}]_0^1 \\ &= -2(e^0 - e^1) \\ &= 2(e-1)\end{aligned}$$

b)

$$\begin{aligned}y &= \sin^{-1}(3x-2) \\ \frac{dy}{dx} &= \frac{3}{\sqrt{1-(3x-2)^2}} \\ \text{at } x &= \frac{1}{2} \\ \frac{dy}{dx} &= \frac{3}{\sqrt{1-\left(\frac{3}{2}-2\right)^2}} \\ &= \frac{3}{\sqrt{1-\frac{1}{4}}} \\ &= 2\sqrt{3}\end{aligned}$$

∴ the gradient of the tangent to the graph

$$y = \sin^{-1}(3x-2) \text{ at } x = \frac{1}{2} \text{ is } 2\sqrt{3}$$

c) Standard Integral sheet:

$$\int \sec x \tan x \, dx = \sec x$$

$$\therefore \sec x \tan x = \frac{d}{dx} \sec x$$

$$\text{Now, } y = \log_e(\sec x)$$

$$\therefore \frac{dy}{dx} = \frac{\sec x \tan x}{\sec x}$$

$$\therefore \frac{dy}{dx} = \tan x$$

d)

$$\begin{aligned}\cos^{-1}\left(\sin\left(-\frac{\pi}{6}\right)\right) &= \cos^{-1}\left(-\sin\frac{\pi}{6}\right) \\ &= \pi - \cos^{-1}\left(\sin\frac{\pi}{6}\right) \\ &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3}\end{aligned}$$

e) Let $x = \cos \theta$

$$dx = -\sin \theta d\theta, \text{ where } 0 < \theta < \pi.$$

$$\text{For } x = \frac{1}{2}, \theta = \frac{\pi}{3} \text{ and for } x = 0, \theta = \frac{\pi}{2}$$

$$\begin{aligned}&\int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \frac{\cos \theta}{\sqrt{1-\cos^2 \theta}} \times (-\sin \theta) d\theta \\ &= \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \frac{\cos \theta}{\sin \theta} \times (-\sin \theta) d\theta \\ &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos \theta \, d\theta \\ &= \left[\sin \theta \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} \\ &= \sin \frac{\pi}{2} - \sin \frac{\pi}{3} \\ &= 1 - \frac{\sqrt{3}}{2}\end{aligned}$$

Question 7

a) Let $A = \tan^{-1} \frac{1}{5}$ then

$$\begin{aligned}\tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \\ &= \frac{2 \times \frac{1}{5}}{1 - \frac{1}{25}} \\ &= \frac{5}{12}\end{aligned}$$

$$\therefore \tan^{-1} \frac{5}{12} = 2A$$

$$\therefore \tan^{-1} \frac{5}{12} = 2 \tan^{-1} \frac{1}{5}$$

b)

(i)

$$\begin{aligned}f(x) &= 1 - e^{-x^2} \\ f'(x) &= 2xe^{-x^2} \\ f''(x) &= 2e^{-x^2} + 2x \times (-2x)e^{-x^2} \\ &= 2e^{-x^2} (1 - 2x^2) \\ f''(x) &= 0 \\ \therefore 2e^{-x^2} (1 - 2x^2) &= 0 \\ \therefore x &= -\frac{1}{\sqrt{2}} \quad \text{as } e^{-x^2} \neq 0 \\ \therefore f\left(-\frac{1}{\sqrt{2}}\right) &= 1 - e^{-\frac{1}{2}} \\ \therefore y &= 1 - \frac{1}{\sqrt{e}}\end{aligned}$$

x	-1	$-\frac{1}{\sqrt{2}}$	0
$f''(x)$	-0.36	0	2

Concavity changes

$\therefore$ the point of inflexion is

$$\left(-\frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{e}}\right)$$

(ii)

As $x \rightarrow \infty, e^{-x^2} \rightarrow 0$ therefore

$$1 - e^{-x^2} \rightarrow 1 \quad \therefore f(x) \rightarrow 1$$

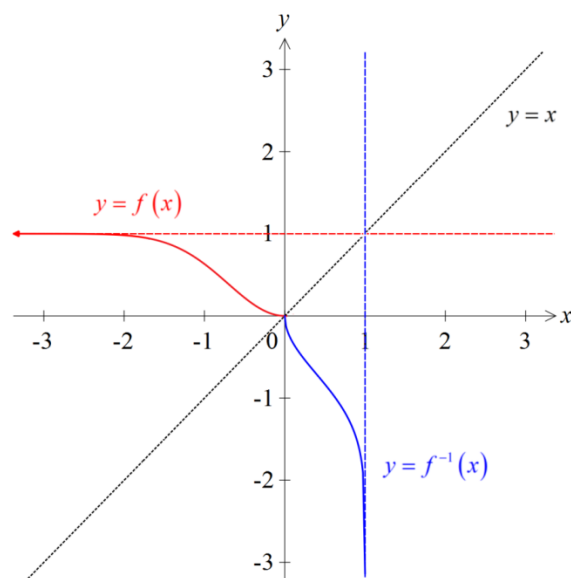
Min turning point is at $(0, 0)$

$$\therefore f(x) \geq 0$$

Hence $0 \leq f(x) < 1$

$$(iii) \quad \lim_{x \rightarrow -\infty} f'(x) = 0$$

(iv)



(v) As above

(vi)

$$\begin{aligned}x &= 1 - e^{-y^2} \\ e^{-y^2} &= 1 - x \\ -y^2 &= \log_e (1 - x) \\ y^2 &= \log_e \frac{1}{1 - x}\end{aligned}$$

as $y \leq 0$

$$\therefore y = -\sqrt{\log_e \frac{1}{1 - x}} \quad \text{where } 0 \leq x \leq 1$$

Question 8 (12 Marks)

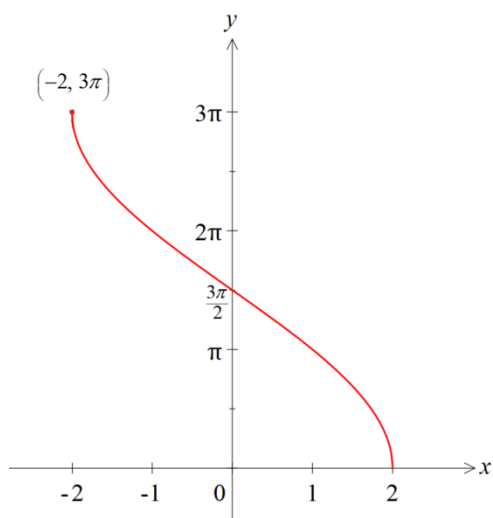
a)

(i)

$$\begin{aligned} f(x) &= 3 \cos^{-1} \frac{0}{2} \\ &= 3 \times \frac{\pi}{2} \\ &= \frac{3\pi}{2} \end{aligned}$$

$\therefore$ The y intercept of the graph of the function $y = f(x)$ is $\frac{3\pi}{2}$

(ii)



(iii)

$$y = 3 \cos^{-1} \frac{x}{2}$$

$$\frac{y}{3} = \cos^{-1} \frac{x}{2}$$

$$\frac{x}{2} = \cos \frac{y}{3}$$

$$\therefore x = 2 \cos \frac{y}{3}$$

$$\begin{aligned} V &= \pi \int_0^\pi x^2 dy - \pi \times 1^2 \times \pi \\ &= \pi \int_0^\pi 4 \cos^2 \frac{y}{3} dy - \pi^2 \\ &= 2\pi \int_0^\pi \left(\cos \frac{2y}{3} + 1 \right) dy - \pi^2 \\ &= 2\pi \left[\frac{3}{2} \sin \frac{2y}{3} + y \right]_0^\pi - \pi^2 \\ &= 2\pi \left(\frac{3}{2} \sin \frac{2\pi}{3} + \pi - 0 \right) - \pi^2 \\ &= 3\pi \times \frac{\sqrt{3}}{2} + 2\pi^2 - \pi^2 \\ &= \frac{3\pi\sqrt{3}}{2} + \pi^2 \end{aligned}$$

$$\therefore V = \pi \left(\frac{3\sqrt{3}}{2} + \pi \right) \text{ u}^3$$

b)

(i)

$$y = \log_e (x+1)$$

$$\therefore x = e^y - 1$$

$$A = k \times \log_e (k+1) - \int_0^{\log_e (k+1)} (e^y - 1) dy$$

$$= k \log_e (k+1) - [e^y - y]_0^{\log_e (k+1)}$$

$$= k \log_e (k+1) - (k+1 - \log_e (k+1) - 1)$$

$$= k \log_e (k+1) - (k - \log_e (k+1))$$

$$\therefore A = (k+1) \log_e (k+1) - k$$

(ii)

$$\log_e (k+1) > \frac{2k}{k+2} \text{ for } k > 0$$

Area of triangle is

$$\frac{k}{2} \log_e (k+1) \text{ where } k > 0$$

Area of the region > Area of the triangle

$$\therefore (k+1) \log_e (k+1) - k > \frac{k}{2} \log_e (k+1)$$

$$2(k+1) \log_e (k+1) - 2k > k \log_e (k+1)$$

$$(2k+2-k) \log_e (k+1) > 2k$$

$$\therefore \log_e (k+1) > \frac{2k}{k+2} \text{ for } k > 0$$

(iii)

$$\text{For } x = k+1 \rightarrow k = x-1$$

$$\text{For } k > 0$$

$$x-1 > 0$$

$$x > 1$$

$$\therefore \log_e (x-1+1) > \frac{2(x-1)}{x-1+2}$$

$$\log_e x > \frac{2(x-1)}{x+1}$$

$$\text{Hence } \frac{\log_e x}{2} > \frac{(x-1)}{x+1} \text{ for } x > 1$$